

Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	1. Summary The company experienced a Distributed Denial of Service (DDoS) attack that disrupted its internal network for two hours. The attacker sent a large number of ICMP packets (ping requests) through a firewall that was not properly configured. This overwhelmed the network and prevented employees from accessing network resources. The incident management team responded by blocking incoming ICMP packets, shutting down non-critical services, and restoring important network services. After investigating, the cybersecurity team introduced new security measures to prevent similar attacks.
Identify	The attack was a Distributed Denial of Service (DDoS) attack, specifically an ICMP flood attack. The attack affected the company's internal network and prevented employees from accessing network resources and services. The main security weakness was an improperly configured firewall that allowed too many ICMP packets into the network. To prevent future incidents, the company should regularly check its network devices, firewall settings, and access permissions to identify security weaknesses.

Protect	To protect the company's network from future attacks, the following steps should be taken: Configure the firewall to limit the number of incoming ICMP packets. Enable IP address verification to help identify fake or spoofed IP addresses. Regularly update network devices and security software. Use an IDS/IPS system to filter suspicious network traffic. Train employees on basic cybersecurity practices and how to report suspicious activities. These measures will help reduce the chances of another successful attack**.**
Detect	The company should continuously monitor its network to detect unusual activities before they cause serious damage. Network monitoring software can help identify sudden increases in traffic, especially large numbers of ICMP packets. An Intrusion Detection System (IDS) can alert the security team when suspicious traffic is detected, while an Intrusion Prevention System (IPS) can help block harmful traffic. The cybersecurity team should also regularly review firewall logs, network activity, and user login records to identify unauthorized access attempts or unusual activities.
Respond	If another DDoS attack occurs, the cybersecurity team should respond quickly to reduce its impact. First, the team should identify the suspicious traffic and block or limit it using firewall rules and the IDS/IPS system. Next, non-critical network services can be temporarily disabled to reduce network pressure while important services are protected. The team should investigate the incident by reviewing firewall logs, network traffic records, and security alerts to understand how the attack happened. After resolving the issue, the incident should be documented, and the security team should update its response procedures to improve how future attacks are handled.

Recover	After stopping the attack, the company should restore all affected network services and confirm that they are working properly. The cybersecurity team should test network connections and check that employees can access the resources they need. The team should also check for any remaining security weaknesses and ensure that firewall rules and monitoring tools are functioning correctly. Important company data should be backed up regularly in case future incidents affect data availability. Finally, the company should review what happened during the attack and improve its recovery plan to reduce downtime during future incidents.

Reflections/Notes: This activity helped me understand how a DDoS attack can affect a company's network and why proper firewall configuration is important. I also learned that cybersecurity is not only about stopping attacks but also about identifying risks, protecting systems, detecting threats, responding quickly, and recovering from incidents. Using the NIST Cybersecurity Framework can help organizations prepare for security threats and improve their overall network security.